

MOBİL PROGRAMLAMA

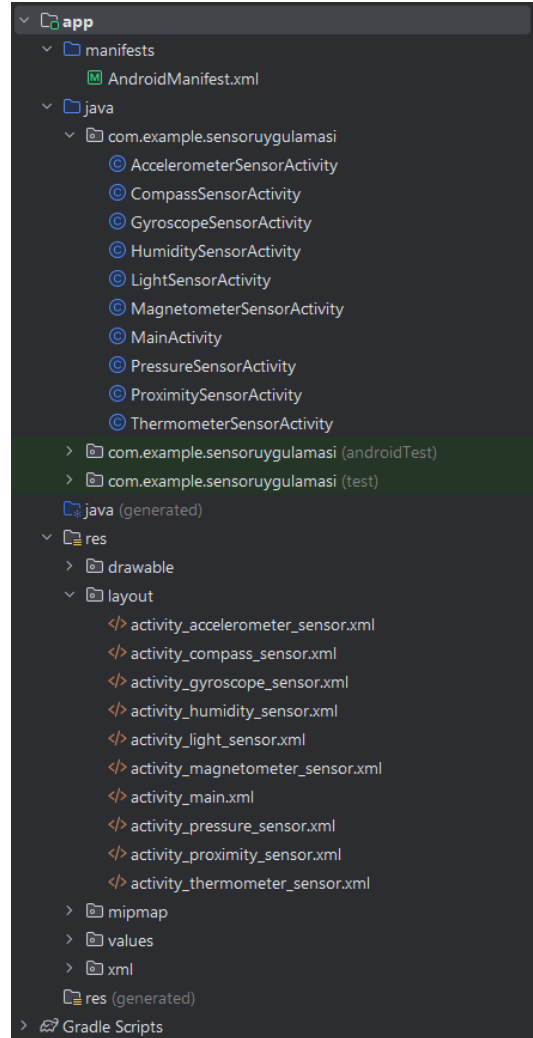
Öğr. Gör. Fatih Kazdal

Sensör Uygulaması

- Sensörlerin işlevlerini görmek için bir uygulama tasarlayacağız.
- Bu uygulama, bize sensörlerin nasıl kullanıldığı hakkında bilgi verecek.
- Android Studio'da "**Empty Views Activity**" şablonuyla yeni bir proje oluşturun.
- Toplam 9 sensör kullanacağımız için 9 Adet **Activity** oluşturun.
- Ardından aşağıda verilen adımları takip edebilirsiniz.

Sensör Uygulaması

(Proje Dizini)



Sensör Uygulaması

(Main Activity Tasarım)

Accelerometer Sensor

Compass Sensor

Gyroscope Sensor

Humidity Sensor

Light Sensor

Magnetometer Sensor

Pressure Sensor

Proximity Sensor

Thermometer Sensor

Button

Button

Button

Button

Button

Button

Button

Button

Button

Sensör Uygulaması

(Accelerometer Sensor Activity Tasarım)

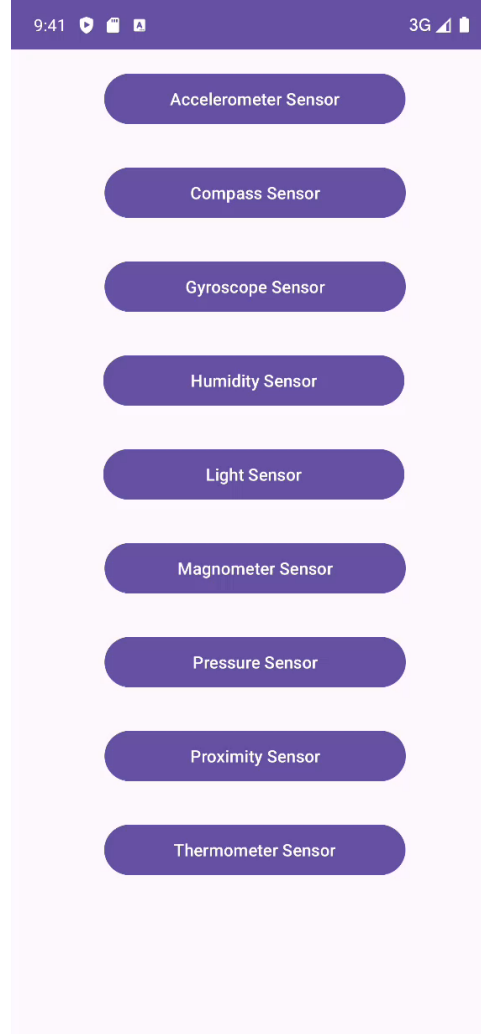
Accelerometer Data



Diğer tüm sensör ekranı tasarımları bu şekilde olacaktır.

Sensör Uygulaması

(Proje Videosu)



Sensör Uygulaması

(app/manifests/AndroidManifest.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SensorUygulaması"
        tools:targetApi="31">
        <activity
            android:name=".MagnetometerSensorActivity"
            android:exported="false" />
        <activity
            android:name=".HumiditySensorActivity"
            android:exported="false" />
        <activity
            android:name=".ThermometerSensorActivity"
            android:exported="false" />
        <activity
            android:name=".AccelerometerSensorActivity"
            android:exported="false" />
        <activity
            android:name=".PressureSensorActivity"
            android:exported="false" />
        <activity
            android:name=".GyroscopeSensorActivity"
            android:exported="false" />
        <activity
            android:name=".CompassSensorActivity"
            android:exported="false" />
        <activityv
```

Sensör Uygulaması

(app/manifests/AndroidManifest.xml)

```
<activity
    android:name=".CompassSensorActivity"
    android:exported="false" />

<activity
    android:name=".ProximitySensorActivity"
    android:exported="false" />

<activity
    android:name=".LightSensorActivity"
    android:exported="false" />

<activity
    android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>

</manifest>
```


Sensör Uygulaması

(app/res/layout/activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button2"
        android:layout_width="wrap_content"
        android:layout_height="80dp"
        android:layout_marginEnd="3dp"
        android:layout_marginBottom="19dp"
        android:text="Button"
        app:layout_constraintBottom_toTopOf="@+id/sensor1_button"
        app:layout_constraintEnd_toEndOf="@+id/sensor1_button" />

    <Button
        android:id="@+id/sensor1_button"
        android:layout_width="250dp"
        android:layout_height="50dp"
        android:layout_marginTop="16dp"
        android:text="Accelerometer Sensor"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/sensor6_button"
        android:layout_width="250dp"
        android:layout_height="50dp"
        android:layout_marginTop="28dp"
        android:text="Magnetometer Sensor"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.503"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/sensor5_button" />

</Button>
```

Sensör Uygulaması

(app/res/layout/activity_main.xml)

```
<Button
    android:id="@+id/sensor3_button"
    android:layout_width="250dp"
    android:layout_height="50dp"
    android:layout_marginTop="28dp"
    android:text="Gyroscope Sensor"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.503"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/sensor2_button" />

<Button
    android:id="@+id/sensor4_button"
    android:layout_width="250dp"
    android:layout_height="50dp"
    android:layout_marginTop="28dp"
    android:text="Humidity Sensor"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.496"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/sensor3_button" />

<Button
    android:id="@+id/sensor5_button"
    android:layout_width="250dp"
    android:layout_height="50dp"
    android:layout_marginTop="28dp"
    android:text="Light Sensor"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.496"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/sensor4_button" />

<Button
    android:id="@+id/sensor2_button"
    android:layout_width="250dp"
    android:layout_height="50dp"
    android:layout_marginTop="28dp"
    android:text="Compass Sensor"
    app:layout_constraintEnd_toEndOf="parent"
```

Sensör Uygulaması

(app/res/layout/activity_main.xml)

```
        android:text="Compass_Sensor"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.503"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/sensor1_button" />
    <Button
        android:id="@+id/sensor7_button"
        android:layout_width="250dp"
        android:layout_height="50dp"
        android:layout_marginTop="28dp"
        android:text="Pressure_Sensor"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.503"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/sensor6_button" />
    <Button
        android:id="@+id/sensor8_button"
        android:layout_width="250dp"
        android:layout_height="50dp"
        android:layout_marginTop="28dp"
        android:text="Proximity_Sensor"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.503"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/sensor7_button" />
    <Button
        android:id="@+id/sensor9_button"
        android:layout_width="250dp"
        android:layout_height="50dp"
        android:layout_marginTop="28dp"
        android:text="Thermometer_Sensor"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/sensor8_button" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Sensör Uygulaması

(app/res/layout/activity_accelometer_sensor.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center">

    <TextView
        android:id="@+id/accelerometer_text_view"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="24sp"
        android:text="Accelerometer Data"
        android:layout_margin="16dp"/>

</LinearLayout>
```

Diğer tüm sensör ekranı kodları bu şekilde olacaktır.

Sensör Uygulaması

(app/java/MainActivity.java)

```
package com.example.sensoruygulamasi;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button sensor1Button = findViewById(R.id.sensor1_button);
        sensor1Button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startActivity(new Intent( packageContext: MainActivity.this, AccelerometerSensorActivity.class));
            }
        });

        Button sensor2Button = findViewById(R.id.sensor2_button);
        sensor2Button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startActivity(new Intent( packageContext: MainActivity.this, CompassSensorActivity.class));
            }
        });

        Button sensor3Button = findViewById(R.id.sensor3_button);
        sensor3Button.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startActivity(new Intent( packageContext: MainActivity.this, GyroscopeSensorActivity.class));
            }
        });
    }
}
```

Sensör Uygulaması

(app/java/MainActivity.java)

```
Button sensor3Button = findViewById(R.id.sensor3_button);
sensor3Button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        startActivity(new Intent( packageContext: MainActivity.this, GyroscopeSensorActivity.class));
    }
});

Button sensor4Button = findViewById(R.id.sensor4_button);
sensor4Button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        startActivity(new Intent( packageContext: MainActivity.this, HumiditySensorActivity.class));
    }
});

Button sensor5Button = findViewById(R.id.sensor5_button);
sensor5Button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        startActivity(new Intent( packageContext: MainActivity.this, LightSensorActivity.class));
    }
});

Button sensor6Button = findViewById(R.id.sensor6_button);
sensor6Button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        startActivity(new Intent( packageContext: MainActivity.this, MagnetometerSensorActivity.class));
    }
});

Button sensor7Button = findViewById(R.id.sensor7_button);
sensor7Button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        startActivity(new Intent( packageContext: MainActivity.this, PressureSensorActivity.class));
    }
});
```

Sensör Uygulaması

(app/java/MainActivity.java)

```
Button sensor7Button = findViewById(R.id.sensor7_button);
sensor7Button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        startActivity(new Intent( packageContext: MainActivity.this, PressureSensorActivity.class));
    }
});

Button sensor8Button = findViewById(R.id.sensor8_button);
sensor8Button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        startActivity(new Intent( packageContext: MainActivity.this, ProximitySensorActivity.class));
    }
});

Button sensor9Button = findViewById(R.id.sensor9_button);
sensor9Button.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        startActivity(new Intent( packageContext: MainActivity.this, ThermometerSensorActivity.class));
    }
});
}
```

Sensör Uygulaması

(app/java/AccelerometerSensorActivity.java)

```
package com.example.sensoruygulamasi;

import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.os.Bundle;
import android.view.MenuItem;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class AccelerometerSensorActivity extends AppCompatActivity implements SensorEventListener {

    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor accelerometerSensor;
    2 usages
    private TextView accelerometerTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_accelerometer_sensor);

        getSupportActionBar().setDisplayHomeAsUpEnabled(true);

        accelerometerTextView = findViewById(R.id.accelerometer_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        accelerometerSensor = sensorManager.getDefaultSensor(Sensor.TYPE_ACCELEROMETER);
    }

    @Override
    protected void onResume() {
        super.onResume();
        sensorManager.registerListener(listener: this, accelerometerSensor, SensorManager.SENSOR_DELAY_NORMAL);
    }
}
```


Sensör Uygulaması

(app/java/AccelerometerSensorActivity.java)

```
@Override
protected void onResume() {
    super.onResume();
    sensorManager.registerListener(listener: this, accelerometerSensor, SensorManager.SENSOR_DELAY_NORMAL);
}

@Override
protected void onPause() {
    super.onPause();
    sensorManager.unregisterListener(this);
}

@Override
public void onSensorChanged(SensorEvent event) {
    float xValue = event.values[0];
    float yValue = event.values[1];
    float zValue = event.values[2];
    accelerometerTextView.setText("X: " + xValue + "\nY: " + yValue + "\nZ: " + zValue);
}

@Override
public void onAccuracyChanged(Sensor sensor, int accuracy) {

}

@Override
public boolean onOptionsItemSelected(MenuItem item) {
    if (item.getItemId() == android.R.id.home) {
        finish();
        return true;
    }
    return super.onOptionsItemSelected(item);
}
```

Sensör Uygulaması

(app/java/CompassSensorActivity.java)

```
package com.example.sensoruygulamasi;

import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class CompassSensorActivity extends AppCompatActivity implements SensorEventListener {

    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor compassSensor;
    2 usages
    private TextView compassTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_compass_sensor);

        compassTextView = findViewById(R.id.compass_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        compassSensor = sensorManager.getDefaultSensor(Sensor.TYPE_MAGNETIC_FIELD);
    }

    @Override
    protected void onResume() {
        super.onResume();
        sensorManager.registerListener(listener: this, compassSensor, SensorManager.SENSOR_DELAY_NORMAL);
    }
}
```

Sensör Uygulaması

(app/java/CompassSensorActivity.java)

```
@Override
protected void onResume() {
    super.onResume();
    sensorManager.registerListener(listener: this, compassSensor, SensorManager.SENSOR_DELAY_NORMAL);
}

@Override
protected void onPause() {
    super.onPause();
    sensorManager.unregisterListener(this);
}

@Override
public void onSensorChanged(SensorEvent event) {
    float compassValue = event.values[0];
    compassTextView.setText("Compass: " + compassValue);
}

@Override
public void onAccuracyChanged(Sensor sensor, int accuracy) {

}

}
```

Sensör Uygulaması

(app/java/GyroscopeSensorActivity.java)

```
package com.example.sensoruygulamasi;

import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class GyroscopeSensorActivity extends AppCompatActivity implements SensorEventListener {

    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor gyroscopeSensor;
    2 usages
    private TextView gyroscopeTextView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_gyroscope_sensor);

        gyroscopeTextView = findViewById(R.id.gyroscope_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        gyroscopeSensor = sensorManager.getDefaultSensor(Sensor.TYPE_GYROSCOPE);
    }

    @Override
```

Sensör Uygulaması

(app/java/GyroscopeSensorActivity.java)

```
@Override
protected void onResume() {
    super.onResume();
    sensorManager.registerListener(listener: this, gyroscopeSensor, SensorManager.SENSOR_DELAY_NORMAL);
}

@Override
protected void onPause() {
    super.onPause();
    sensorManager.unregisterListener(this);
}

@Override
public void onSensorChanged(SensorEvent event) {
    float xValue = event.values[0];
    float yValue = event.values[1];
    float zValue = event.values[2];
    gyroscopeTextView.setText("X: " + xValue + "\nY: " + yValue + "\nZ: " + zValue);
}

@Override
public void onAccuracyChanged(Sensor sensor, int accuracy) {
}
}
```

Sensör Uygulaması

(app/java/HumiditySensorActivity.java)

```
package com.example.sensoruygulamasi;
import ...

public class HumiditySensorActivity extends AppCompatActivity implements SensorEventListener {
    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor humiditySensor;
    2 usages
    private TextView humidityTextView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_humidity_sensor);

        humidityTextView = findViewById(R.id.humidity_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        humiditySensor = sensorManager.getDefaultSensor(Sensor.TYPE_RELATIVE_HUMIDITY);
    }
    @Override
    protected void onResume() {
        super.onResume();
        sensorManager.registerListener(listener this, humiditySensor, SensorManager.SENSOR_DELAY_NORMAL);
    }
    @Override
    protected void onPause() {
        super.onPause();
        sensorManager.unregisterListener(this);
    }
    @Override
    public void onSensorChanged(SensorEvent event) {
        float humidityValue = event.values[0];
        humidityTextView.setText("Humidity: " + humidityValue);
    }
    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {
    }
}
```

Sensör Uygulaması

(app/java/LightSensorActivity.java)

```
package com.example.sensoruygulamasi;
import ...

public class LightSensorActivity extends AppCompatActivity implements SensorEventListener {
    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor lightSensor;
    2 usages
    private TextView lightTextView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_light_sensor);

        lightTextView = findViewById(R.id.light_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        lightSensor = sensorManager.getDefaultSensor(Sensor.TYPE_LIGHT);
    }
    @Override
    protected void onResume() {
        super.onResume();
        sensorManager.registerListener(listener: this, lightSensor, SensorManager.SENSOR_DELAY_NORMAL);
    }
    @Override
    protected void onPause() {
        super.onPause();
        sensorManager.unregisterListener(this);
    }
    @Override
    public void onSensorChanged(SensorEvent event) {
        float lightValue = event.values[0];
        lightTextView.setText("Light: " + lightValue);
    }
    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {
    }
}
```

Sensör Uygulaması

(app/java/MagnetometerSensorActivity.java)

```
package com.example.sensoruygulamasi;
import ...
public class MagnetometerSensorActivity extends AppCompatActivity implements SensorEventListener {
    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor magnetometerSensor;
    2 usages
    private TextView magnetometerTextView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_magnetometer_sensor);
        magnetometerTextView = findViewById(R.id.magnetometer_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        magnetometerSensor = sensorManager.getDefaultSensor(Sensor.TYPE_MAGNETIC_FIELD);
    }
    @Override
    protected void onResume() {
        super.onResume();
        sensorManager.registerListener(listener: this, magnetometerSensor, SensorManager.SENSOR_DELAY_NORMAL);
    }
    @Override
    protected void onPause() {
        super.onPause();
        sensorManager.unregisterListener(this);
    }
    @Override
    public void onSensorChanged(SensorEvent event) {
        float xValue = event.values[0];
        float yValue = event.values[1];
        float zValue = event.values[2];
        magnetometerTextView.setText("X: " + xValue + "\nY: " + yValue + "\nZ: " + zValue);
    }
    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {
    }
}
```


Sensör Uygulaması

(app/java/PressureSensorActivity.java)

```
package com.example.sensoruygulamasi;
import ...

public class PressureSensorActivity extends AppCompatActivity implements SensorEventListener {
    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor pressureSensor;
    2 usages
    private TextView pressureTextView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_pressure_sensor);

        pressureTextView = findViewById(R.id.pressure_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        pressureSensor = sensorManager.getDefaultSensor(Sensor.TYPE_PRESSURE);
    }
    @Override
    protected void onResume() {
        super.onResume();
        sensorManager.registerListener(listener: this, pressureSensor, SensorManager.SENSOR_DELAY_NORMAL);
    }
    @Override
    protected void onPause() {
        super.onPause();
        sensorManager.unregisterListener(this);
    }
    @Override
    public void onSensorChanged(SensorEvent event) {
        float pressureValue = event.values[0];
        pressureTextView.setText("Pressure: " + pressureValue);
    }
    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {
    }
}
```

Sensör Uygulaması

(app/java/ProximitySensorActivity.java)

```
package com.example.sensoruygulamasi;
import ...

public class ProximitySensorActivity extends AppCompatActivity implements SensorEventListener {
    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor proximitySensor;
    2 usages
    private TextView proximityTextView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_proximity_sensor);

        proximityTextView = findViewById(R.id.proximity_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        proximitySensor = sensorManager.getDefaultSensor(Sensor.TYPE_PROXIMITY);
    }
    @Override
    protected void onResume() {
        super.onResume();
        sensorManager.registerListener(listener: this, proximitySensor, SensorManager.SENSOR_DELAY_NORMAL);
    }
    @Override
    protected void onPause() {
        super.onPause();
        sensorManager.unregisterListener(this);
    }
    @Override
    public void onSensorChanged(SensorEvent event) {
        float proximityValue = event.values[0];
        proximityTextView.setText("Proximity: " + proximityValue);
    }
    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {
    }
}
```

Sensör Uygulaması

(app/java/ThermometerSensorActivity.java)

```
package com.example.sensoruygulamasi;
import ...

public class ThermometerSensorActivity extends AppCompatActivity implements SensorEventListener {
    4 usages
    private SensorManager sensorManager;
    2 usages
    private Sensor thermometerSensor;
    2 usages
    private TextView thermometerTextView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_thermometer_sensor);

        thermometerTextView = findViewById(R.id.thermometer_text_view);
        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
        thermometerSensor = sensorManager.getDefaultSensor(Sensor.TYPE_AMBIENT_TEMPERATURE);
    }
    @Override
    protected void onResume() {
        super.onResume();
        sensorManager.registerListener(listener: this, thermometerSensor, SensorManager.SENSOR_DELAY_NORMAL);
    }
    @Override
    protected void onPause() {
        super.onPause();
        sensorManager.unregisterListener(this);
    }
    @Override
    public void onSensorChanged(SensorEvent event) {
        float temperatureValue = event.values[0];
        thermometerTextView.setText("Temperature: " + temperatureValue);
    }
    @Override
    public void onAccuracyChanged(Sensor sensor, int accuracy) {
    }
}
```